Junyan Liu

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Education

Columbia University in the City of New York — New York, US
MS in Mechanical Engineering (Robotics and Control Track) — *Sep 2025 – Current*
- Coursework: Robotic Studio (Advanced), Robot Learning, Model Predictive Control, Introduction to Robotics

University of Wollongong — Wollongong, AU
BE in Mechatronic Engineering (Honours) — *Jul 2024 – Jun 2025*
- Coursework: Applied Topics in Mechatronics, Reliability Engineering, Manufacturing Engineering Principles

Beijing Jiaotong University — Beijing, CN
BE in Mechatronic Engineering — *Sep 2021 – Jun 2025*
- Coursework: Electronics, Programming for Engineers, Embedded System and Applications

Honors and Awards

Winner, Best Use of Computer Vision — MakeCU 2025 Hackathon (Columbia Robotics Club) — Nov 2025

Research Experience

World Action Model for Robotic Manipulation — Jun 2026 – Present
RoboPIL Lab, Prof. Yunzhu Li (mentored by PhD student Kaifeng Zhang)
- Explored MPPI baselines (Franka grasping in Newton, task planning in RoboCasa), then shifted to learned World Action Models (WAMs) built on the Interactive World Simulator (IWS).
- On Push-T, benchmarked WAM variants built on frozen IWS dynamics against Diffusion Policy and a 450M SmolVLA; the best matched SmolVLA (171 vs. 172 of 250 episodes).
- Scaled to RoboDojo (Isaac Sim): built an automated data-collection pipeline (11.9M frames, 31k episodes) and trained IWS from scratch on this self-collected data.
- Reproduced the VPP2 baseline (14B video + 2B action) within leaderboard noise on three tasks; now training per-task WAMs (IWS → video DiT → action DiT) to compare against it.

Replicate Project: Self-Reproducing Modular Robots — Feb 2026 – Present
Creative Machines Lab, Prof. Hod Lipson
- Co-developed the control stack for a 15-module self-reproducing robot: ESP32 modules execute ESP-NOW-synchronized locomotion, module-carrying, and budding sequences across 5-module "seal" and 3-module "trilobite" morphologies.
- Built a closed-loop vision system (OpenCV, ArUco) that tracks module poses and steers the robot to dock, achieving fully autonomous collection of all scattered modules in physical trials.
- Engineered a custom magnetic physics engine for MuJoCo ($O(N^2)$ model with distance attenuation and damping) whose docking-force parameters were adopted for the physical docking interface.
- Migrated simulation to MuJoCo/MJX and wrote a C library injecting forces directly into MuJoCo's low-level arrays, achieving a 2.34× per-step speedup for large-scale evolutionary search.

Influence of Magnet Internal Structure on Magnetic Performance — Aug 2024 – Jun 2025
University of Wollongong, Dr. Chin-Hsing Kuo
- Investigated how internal hollow structures affect magnetic force in cylindrical and rectangular NdFeB magnets via dual-platform simulation: MATLAB (Monte Carlo dipole-dipole modeling with 1,000,000+ sampling points per configuration) and ANSYS Maxwell 2024 (finite element analysis using the Virtual Work method).
- Developed a custom MATLAB simulation pipeline (dipole-dipole interaction model, systematic geometry sweeps, cubic polynomial curve fitting) to extract force-geometry relationships, cross-validated against ANSYS Maxwell FEM results.
- Identified a non-linear, cubic relationship between internal hollow size and magnetic force; found rectangular magnets exhibit higher initial force but a steeper decline than cylindrical magnets, attributable to edge and corner flux-distortion effects.
- Presented research findings to academic supervisors and peers, communicating technical results to both engineering and non-technical audiences.

Legged Robot Learns to Walk via Reinforcement Learning — Jan 2024 – Feb 2024
Global Education, Academics, and Research Skills (GEARS)
- Configured a Ubuntu-based RL simulation environment and designed reward functions to train a legged robot controller, evaluating walking performance across different reward compositions.

Publications

S. Zhang*, **J. Liu***, W. He*, H. Lipson, "Robotic Budding: From Homogeneous Cells to Diverse Morphologies in Physical Self-Reproduction"
IROS 2026 Workshop on Challenges and Applications Prospects for Reconfigurable Modular Robots, 2026.
IROS 2026 Workshop on Bridging the Gap between Robotic Matter and Active Metamaterials, 2026.
*Equal contribution.

Teaching Experience

Columbia University New York, US
Teaching Assistant *Sep 2026 – Present*
MECE4058E: Mechatronics & Embedded Micro *Instructor: Enrico Zordan*

- Helped students understand and complete lab exercises in embedded microcontroller programming (Assembly/C) and mechatronic systems (solenoids, stepper/DC motors, thermal systems, magnetic levitation), and graded lab results.

Internship Experience

C-Based Structural Simulation Software Developer Nov 2024 – Feb 2025
Shenzhen Yunbo Software Technology Co., Ltd.

- Developed and optimized structural simulation software in C, enhancing performance of model setup, simulation core, and result visualization modules; improved simulation efficiency by 20%.
- Collaborated with a cross-functional team of 7 engineers to identify requirements and optimize algorithms, ensuring accuracy within 2% deviation from benchmark data.
- Conducted 50+ test/debug cycles, reducing runtime errors by 30% and increasing software stability.

Projects Design

Robotic Studio: Quadruped Robot "Pack-Pack" Development Sep 2025 – Dec 2025

- Designed and developed a custom quadruped robot from concept sketches to physical deployment, completing the entire pipeline of mechanical design, simulation, and control.
- Engineered the mechanical structure using SolidWorks for 3D CAD modeling and assembled the physical prototype integrating a Raspberry Pi core controller and 8 servo motors (2 DOF per leg).
- Constructed a high-fidelity physics simulation environment utilizing MuJoCo to safely and efficiently design, test, and evaluate various locomotion control strategies.
- Implemented and evaluated multiple control algorithms within the simulation, including preset gaits, Random Search, Hill Climbing, and Reinforcement Learning (RL), analyzing models optimized for maximum speed and gait stability.
- Successfully achieved sim-to-real transfer of heuristic search algorithms, deploying locomotion policies optimized via Random Search and Hill Climbing onto the physical hardware to achieve a stable real-world walking speed of 26.47 cm/s.

Google It: Computer-Vision Tracking Robot (MakeCU 2025 Hackathon) Nov 2025

- Built an autonomous mobile robot with a 4-person team in a 24-hour hackathon; pivoted from an initial two-wheel self-balancing design (PID control via MPU6050 IMU) to a three-wheel, camera-guided tracking robot after balance tuning proved infeasible within the time limit.
- Implemented real-time ArUco marker detection and tracking on a Raspberry Pi using OpenCV, computing proportional steering corrections from the marker's pixel offset to drive LX-16A bus-servo-actuated wheels toward the target.
- Integrated an HC-SR04 ultrasonic sensor for obstacle avoidance and an LED status indicator, halting the robot when an obstacle was detected or the target was lost.
- Diagnosed and fixed hardware failures under time pressure — burned-out motors, slipping wheels (resolved with paper shims and wire-wrapped tires), and insufficient battery current (upgraded from 9V to 12V with a DC-DC converter).

Lumbar Surgery Intelligent Planning System May 2023 – May 2024

- Developed an intelligent planning system for lumbar surgery based on X-ray images.
- Utilized deep learning algorithms (U-Net) for automatic lumbar parameter measurements and combined previous surgical data for intelligent surgery planning.
- Aimed to improve surgery efficiency and accuracy, reducing risks through intelligent decision-making support; collaborated with a team under the supervision of Prof. Tian Lixia.

Mechatronics Engineering Cup – Smart Car Challenge Mar 2023 – Jul 2023

- Built an STM32F103VCT-based smart car integrating infrared, ultrasonic, flame, photoresistor, and color sensors to achieve obstacle avoidance, line tracking, fire extinguishing, color-based sorting, and light-source-seeking behaviors.

- Implemented the embedded control stack: PWM motor control, ADC/DAC signal processing, SPI/I2C sensor interfacing, USART communication, and OLED display output.
- Designed individual wheel control logic and integrated it into unified car operation; tested and optimized across multiple terrains to validate reliable multi-function performance.

Ice Mark Grip Oct 2021 – Dec 2021

- Designed a procedure to command the robot's movement in two dimensions and search for objects using sensors.
- Applied hinges, sensors, and basic parts to allow motors to control planes of motion; the robot approaches with water after receiving a signal from the switch, and dispenses water after sensing the object's weight.
- Sensed instructions from the switch: the robot carries water toward the object from a distance and delivers it after serving.

Technical Skills

Programming & Embedded Systems: C/C++, Python, MATLAB, Arduino (ESP32), STM32
Robotics, Simulation & ML: MuJoCo/MJX, Deep Learning (U-Net)
Design & Analysis: SOLIDWORKS, AutoCAD, Inventor, Photoshop, ANSYS Workbench, ANSYS Maxwell
Tools & OS: Linux/Ubuntu, Git